

An Implementation Principle of the Meaning Density Tensor Based on an Ontological Premise Update Operating System

A Unified Theory of Pre-Judgment Determination, Mode of Existence, and Terminal
Choice Structures

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July 4, 2026

Abstract

This study aims to redefine the Premise Update Operating System (Premise Update OS) proposed in Future Convergence Theory (FCT) as an ontological implementation rather than a conventional model of cognitive updating.

Previous studies have characterized premise updating as an operating system responsible for maintaining convergence toward a Future Fixed Point, while suffering has been interpreted as a log indicating the necessity for updating. Although this formulation explains the functional role of updating, it is insufficient for providing an ontological account of states in which the capacity for updating itself is substantially diminished, including depressive states and the emergence of terminal choice.

To address this limitation, the present study adopts the ontological axiom that human beings exist as states of determination prior to judgment. Based on this premise, the concepts of mode of existence, world premises, meaning density, suffering, updating, and terminal choice are reconstructed as components of a single ontological continuum.

Consequently, this study demonstrates that

1. world premises are not cognitive objects but constitute the mode of existence itself;
2. premise updating is not cognitive revision but the updating of one's mode of existence;
3. the Meaning Density Tensor is not a representation of the quantity of meaning but an informational structure that preserves ontological openness (non-closure);
4. suffering is not an ontological failure but an observable log indicating the suspension of world generation; and
5. terminal choice is not the consequence of a single universal cause but one of multiple possible outcomes derived from a particular mode of existence.

Furthermore, by employing examples such as military personnel, religious commitment, social structures, external evaluation, and prolonged social isolation (e.g., NEET states) as counterexamples, this study argues that terminal choice should not be regarded as a universal ontological axiom. Instead, it is more consistently understood as a local consequence emerging from a specific mode of existence.

Finally, the Meaning Density Tensor is redefined as the foundational implementation mechanism of the Ontological Premise Update Operating System. Based on this formulation, a design principle is proposed that maintains world generation itself in an update-capable state without increasing the cognitive burden imposed on human beings.

This study integrates the ontological foundations established in FCT Paper No. 81 and FCT Paper No. 96, thereby extending the Premise Update Operating System from a theory of cognitive updating to a comprehensive Ontological Operating System.

Keywords

Future Convergence Theory (FCT),
Ontology,
Ontological Premise Update Operating System,
Meaning Density Tensor,
Pre-Judgment Determination,
Mode of Existence,
Future Fixed Point,
World Generation,
Suffering,
Terminal Choice,
Depression,
Ontological Openness

1 Introduction

Future Convergence Theory (FCT) is a theoretical framework that seeks to redefine the human being not as a cognitive agent but as an existence that converges toward a Future Fixed Point. Conventional research in cognitive science, psychology, and artificial intelligence has primarily focused on processes that occur after judgment, including decision making, cognition, value formation, and behavioral generation. Although these approaches have produced substantial theoretical and empirical achievements, they generally treat post-judgment information processing as the fundamental object of analysis rather than the ontological condition that precedes judgment itself. FCT reverses this order of explanation. Rather than regarding human beings as subjects who perform judgments, it adopts the position that human beings already exist as states of ontological determination prior to any cognitive judgment. Accordingly, cognition, value, meaning, purpose, self-image, and representations of the future are interpreted as higher-order structures derived from this prior ontological condition. FCT Paper No. 96 introduced this ontological principle under the concept of *Pre-Judgment Determination*, arguing that existence itself constitutes the condition through which a world is already established before cognition begins. Independently, FCT Paper No. 81 formulated the Premise Update Operating System as the operating framework responsible for maintaining convergence toward a Future Fixed Point, while defining suffering as an update request generated within that operating system. When these two theoretical developments are considered together, however, an important ontological question emerges. If existence is already a state of determination prior to judgment, then what exactly is updated through premise updating? Previous interpretations generally regarded premise updating as an operation that modifies cognitive structures, semantic structures, or evaluative frameworks. Within the ontological hierarchy adopted by FCT, however, these structures are themselves consequences of a more fundamental level of existence. If one accepts the hierarchical order

$$\text{Existence} \rightarrow \text{Mode of Existence} \rightarrow \text{World} \rightarrow \text{Meaning} \rightarrow \text{Judgment},$$

then cognitive updating necessarily operates at a higher level than the mode of existence itself. Consequently, cognitive updating and ontological updating cannot be regarded as equivalent operations. The primary objective of this study is therefore to redefine premise updating as the updating of the mode of existence rather than as the modification of cognitive structures. Under this formulation,

- world premises,
- meaning,
- evaluation,
- self-image,
- suffering, and

- update capability

are all understood as structures derived from the underlying mode of existence. A second objective of this study is to reconsider depressive states and terminal choice from an ontological perspective. Contemporary psychology generally explains depression through multiple interacting factors, including affective processes, cognitive mechanisms, neurophysiological conditions, and social environments. Although such explanations are valuable at the phenomenological level, they do not directly address the ontological question of why existence itself can arrive at a condition in which terminal choice becomes possible. Within FCT, terminal choice is not treated as a universal ontological phenomenon. Military service, religious commitment, social roles, external evaluation, and prolonged social isolation represent fundamentally different existential structures, yet each may, under certain circumstances, be associated with terminal choice. Consequently, terminal choice is better understood not as the consequence of a single universal cause but as a local outcome derived from a particular mode of existence. To support this position, the present study employs these examples as counterexamples demonstrating that none of the following constitutes an ontological axiom governing terminal choice:

1. dependence on others,
2. dependence on society,
3. dependence on external evaluation,
4. self-actualization, or
5. loss of meaning.

Instead, this study introduces the concept of the *Mode of Existence* as the fundamental ontological structure that remains after these contingent factors are removed. The Mode of Existence denotes the condition in which a subject already exists as a world prior to recognizing that world through cognition. Here, “world” does not refer to an objective external reality but to the ontological world that has already become established for that particular subject. Accordingly, world premises are not assumptions consciously adopted by an individual; they are constituted by the individual’s own mode of existence. From this standpoint, premise updating does not consist of replacing or modifying meanings. Rather, when the mode of existence itself is updated, the world, meaning, evaluation, and judgment derived from that mode of existence are simultaneously reconstructed as consequences of the ontological update. This study also reinterprets the Meaning Density Tensor. Earlier formulations regarded it primarily as an informational structure for representing the density or quantity of meaning. Once the analysis is extended to the ontological level, however, the Meaning Density Tensor should instead be understood as an informational structure that preserves the ontological openness (non-closure) of the mode of existence. Its essential role is therefore not to quantify meaning but to maintain the possibility of continuous world generation. Through this reinterpretation, the Premise Update Operating System is no longer understood merely as a cognitive updating system but as an Ontological Operating System. Based on these considerations, this paper reconstructs the theoretical foundation of the Ontological Premise Update Operating System and presents the implementation

principle of the Meaning Density Tensor by integrating the ontological foundations established in FCT Paper No. 81 and FCT Paper No. 96.

2 Theoretical Background

This chapter clarifies the theoretical position of the present study. Rather than proposing an entirely new theoretical framework within Future Convergence Theory (FCT), this study integrates the ontological axioms established in previous works and reconstructs them into a unified theory of the Ontological Premise Update Operating System. In particular, FCT Paper No. 81 and FCT Paper No. 96 constitute the immediate theoretical foundation of the present study.

2.1 Relationship to FCT Paper No. 81

FCT Paper No. 81 defined the Premise Update Operating System (Premise Update OS) as the foundational operating framework responsible for maintaining convergence toward a Future Fixed Point. The paper established the following principles:

- Suffering is not an abnormality.
- Suffering functions as an update request indicating divergence from the Future Fixed Point.
- Updating represents the restoration of normal operation.
- Human beings function simultaneously as users of the World OS and as agents responsible for updating it.

Within this framework, suffering is interpreted not as a pathological condition requiring elimination but as information indicating that updating has become necessary. Conventional psychology and psychiatry generally regard suffering as an object of reduction, management, or removal. By contrast, FCT interprets suffering as information generated during the convergence process toward a Future Fixed Point. At the operating-system level, suffering is therefore understood as an update request log. The present study fully retains this interpretation. Suffering remains an update request rather than a pathological phenomenon. However, the present study extends the analysis one ontological level deeper. In FCT Paper No. 81, the primary objects of updating were the World OS and the Personal OS. The present study argues that the World OS itself should be understood as a structure derived from a more fundamental ontological layer, namely the Mode of Existence. Accordingly, the following ontological hierarchy is introduced:

$$\text{Mode of Existence} \rightarrow \text{World OS} \rightarrow \text{Personal OS}.$$

This reconstruction does not replace or contradict the framework established in FCT Paper No. 81. Rather, it provides an ontological foundation beneath the operating-system architecture proposed in that work.

2.2 Relationship to FCT Paper No. 96

FCT Paper No. 96 introduced the ontological axiom that

Existence is already in a state of determination prior to judgment.

This principle differs fundamentally from the conventional view of cognitive science, which assumes an information-processing subject that first perceives and then evaluates the world. Instead, the ontological order adopted by FCT is

$$\text{Existence} \rightarrow \text{World} \rightarrow \text{Judgment},$$

rather than

$$\text{Judgment} \rightarrow \text{World}.$$

Within this framework, meaning, value, purpose, self-image, representations of the future, and cognitive structures are all regarded as higher-order structures derived from existence itself. The present study adopts this ontological axiom without modification. Furthermore, it argues that ontological determination manifests temporally as what is termed the *Mode of Existence*. The Mode of Existence denotes the world that has already become established before the subject recognizes it cognitively. Consequently, the Mode of Existence is not constructed through cognition. Rather, it should be understood as the present manifestation of ontological determination implemented along the temporal dimension.

2.3 Reorganization of the Ontological Hierarchy

When FCT Paper No. 81 and FCT Paper No. 96 are integrated, the theoretical hierarchy can be organized as follows:

$$\text{Existence} \rightarrow \text{Ontological Determination} \rightarrow \text{Mode of Existence} \rightarrow \text{World OS} \rightarrow \text{Personal OS} \rightarrow \text{Cognition} \rightarrow \text{Judgment}$$

Within this hierarchy, cognition is not the foundational layer. The World OS exists beneath cognition, while the Mode of Existence constitutes a more fundamental layer beneath the World OS. Moreover, beneath the Mode of Existence lies Ontological Determination, which is established prior to judgment. Accordingly, the Premise Update Operating System should no longer be interpreted as a cognitive operating system. Instead, it is redefined as the implementation structure of a world derived from Ontological Determination.

2.4 The Position of the Present Study

The present study neither replaces FCT Paper No. 81 nor revises FCT Paper No. 96. Its purpose is instead to complete the missing theoretical layer between the operating-system theory proposed in FCT Paper No. 81 and the ontological axiom established in FCT Paper No. 96. The concept of the Mode of Existence introduced in this study should therefore not be regarded as a new ontological axiom. Rather, it is defined as the present state through which Ontological

Determination unfolds along the temporal dimension. Consequently, the following theoretical sequence is established:

Ontological Determination $\rightarrow$ Mode of Existence $\rightarrow$ World OS $\rightarrow$ Premise Update OS.

Within this framework, the Premise Update Operating System proposed in FCT Paper No. 81 is positioned as an operating system grounded in ontology. Furthermore, the Meaning Density Tensor is redefined not as a representation of cognitive meaning but as an informational structure that preserves the ontological openness maintained by the Mode of Existence. Through this reinterpretation, the following concepts are unified within a single ontological hierarchy:

- Meaning Density Tensor,
- suffering,
- premise updating,
- the Future Fixed Point,
- the World OS,
- the Mode of Existence, and
- Ontological Determination.

This theoretical integration establishes the conceptual foundation of the present study.

3 Ontological Determination and the Mode of Existence

This chapter adopts the concept of *Pre-Judgment Determination*, introduced in FCT Paper No. 96, as a fundamental ontological axiom and clarifies its theoretical relationship to the Mode of Existence. The primary objective of this chapter is to redefine the object updated by the Premise Update Operating System not as a cognitive structure but as an ontological structure.

3.1 The Ontological Position of Pre-Judgment Determination

Conventional cognitive science generally explains world recognition as a process in which a subject perceives the external world, internally processes the acquired information, and subsequently forms judgments. This framework presupposes the following causal sequence:

World $\rightarrow$ Perception $\rightarrow$ Cognition $\rightarrow$ Judgment.

FCT does not adopt this ordering. Instead, FCT regards human beings not as subjects who perform judgments but as beings that already exist in a state of ontological determination prior to judgment. Accordingly, the ontological order becomes

Existence $\rightarrow$ Determination $\rightarrow$ World $\rightarrow$ Cognition $\rightarrow$ Judgment.

Here, the term *determination* does not refer to cognitive judgment. It is independent of logical reasoning, emotional processes, accumulated experience, acquired knowledge, and learning. Rather, it denotes the ontological condition in which existence itself has already been established. Consequently, Pre-Judgment Determination is not an object requiring proof. It functions as an ontological axiom belonging to the same foundational level as existence itself.

3.2 Introducing the Mode of Existence

Although Pre-Judgment Determination establishes the ontological basis of existence, it does not by itself explicitly describe continuity along the temporal dimension. To address this limitation, the present study introduces the concept of the *Mode of Existence*. The Mode of Existence is defined as the present state in which Ontological Determination is continuously implemented through time. A crucial point is that the Mode of Existence is not formed through cognition. Cognition, value, meaning, representations of the future, self-image, and social roles are all structures derived from the Mode of Existence. Accordingly, the Mode of Existence occupies a more fundamental ontological level than any cognitive structure.

3.3 The Mode of Existence and World Premises

Worldviews are commonly understood as systems of values or beliefs consciously adopted by individuals. From an ontological perspective, however, this interpretation is insufficient. A subject does not merely adopt a worldview. Rather, the subject already exists as that world. Accordingly, world premises are not cognitive objects. They are identical to the subject's own Mode of Existence. Religion, nations, societies, communities, self-actualization, and external evaluation should therefore be understood as concrete worlds constituted through particular modes of existence. Consequently, changing a world premise does not mean revising a belief. It means updating the Mode of Existence itself.

3.4 The Mode of Existence and the Process of Formation

The Mode of Existence is not formed simply by accumulating experiences throughout life. Nor is it determined solely by cognitive learning. Nevertheless, it possesses temporal continuity. The present study defines the Mode of Existence as the present manifestation of Ontological Determination unfolding through time. From the perspective of the subject, this temporal unfolding is observed as life experience or as a process of formation. Accordingly, the process of formation should not be regarded as the cause of the Mode of Existence. Rather, it is the observable consequence of the temporal unfolding of the Mode of Existence itself. Under this interpretation, the relationship is not

Mode of Existence $\rightarrow$ Process of Formation,

but instead represents different observational perspectives applied to a single ontological structure:

Ontological Determination = Mode of Existence = Process of Formation.

3.5 Updating the Mode of Existence

Premise updating has traditionally been understood as an operation that revises meanings or cognitive structures. Once the Mode of Existence is introduced, however, cognition is no longer the primary object of updating. Instead, what is updated is the Mode of Existence itself. When the Mode of Existence changes, every higher-order structure derived from it is reconstructed simultaneously. These structures include

- meaning,
- value,
- purpose,
- representations of the future,
- judgment, and
- action.

Accordingly, premise updating is not a sequential modification of individual cognitive elements. Rather, it is an ontological operation that reconstructs the foundational basis from which all such structures are derived.

3.6 The Difference Between Updating the Mode of Existence and Cognitive Updating

Cognitive updating modifies individual items of knowledge, beliefs, or evaluations while assuming that the world itself remains unchanged. Formally, cognitive updating assumes

$$\text{World} = \text{Constant},$$

while cognition alone changes. Ontological updating differs fundamentally. In ontological updating,

$$\text{Mode of Existence} \implies \text{World} \implies \text{Cognition} \implies \text{Judgment}$$

are transformed simultaneously. Consequently, updating the Mode of Existence necessarily includes cognitive updating as one of its consequences. The converse, however, does not hold. Cognitive updating alone cannot entail ontological updating.

3.7 Definition of Ontological Updating

Based on the preceding discussion, the present study defines *Ontological Updating* as follows.

Ontological Updating is the ontological operation in which the Mode of Existence itself is updated, thereby simultaneously reconstructing the world premises, meaning, the Future Fixed Point, values, cognition, judgment, and action that are derived from that Mode of Existence.

Under this definition, the Premise Update Operating System introduced in FCT Paper No. 81 should no longer be interpreted as a Cognitive Update Operating System. Instead, it is more accurately understood as an operating system responsible for updating the Mode of Existence. Furthermore, the concept of Pre-Judgment Determination introduced in FCT Paper No. 96 becomes directly connected to the Premise Update Operating System through the intermediary concept of the Mode of Existence. As a result, ontology, premise updating, meaning density, the Future Fixed Point, and the World OS are unified into a single ontological continuum.

4 Terminal Choice, Depression, and the Limits of Ontological Updating

This chapter examines depression and terminal choice from the perspective of the Ontological Premise Update Operating System. The objective is neither to redefine depression as a medical diagnosis nor to explain terminal choice through a single causal factor. Rather, the purpose is to clarify the ontological structure through which terminal choice may emerge from the relationship between the Mode of Existence and the Premise Update Operating System.

4.1 Terminal Choice Is Not an Ontological Axiom

Terminal choice is often attributed to factors such as depression, loss of meaning, social isolation, external evaluation, or self-denial. However, none of these factors constitutes either a sufficient or a necessary condition. For example, military personnel who prioritize loyalty to the state, individuals who accept self-sacrifice based on religious conviction, members of strongly norm-governed communities, and individuals experiencing prolonged social isolation all possess fundamentally different modes of existence. Nevertheless, under particular circumstances, each may arrive at terminal choice. Conversely, individuals exposed to apparently similar circumstances may never arrive at such a choice. Accordingly, terminal choice should not be understood as the direct consequence of any specific factor. Instead, it is more consistently interpreted as a local consequence derived from a particular Mode of Existence. The present study therefore does not regard terminal choice itself as an ontological axiom.

4.2 The Mode of Existence and World Closure

The Mode of Existence constitutes the world that is already established for a particular subject. Accordingly, world premises are not beliefs consciously adopted by the subject but are identical to the subject's own Mode of Existence. As long as the Mode of Existence preserves ontological updatability, the world remains in a state of ontological openness. When the Mode of Existence undergoes local closure, however, the world becomes observable only as a single possible configuration. The concept of closure employed here does not refer to logical uniqueness. Rather, it denotes a condition in which alternative possibilities for world generation no longer exist ontologically for the subject.

4.3 The Limit of Ontological Updating

FCT Paper No. 81 defined suffering as an update request indicating divergence from the Future Fixed Point. The present study fully preserves this interpretation. Nevertheless, the existence of an update request and the existence of the capability to update are logically distinct. Human beings are influenced by numerous conditions, including neurophysiological factors, cognitive resources, social environments, and accumulated life experiences. As a consequence, when the Mode of Existence converges toward a state of local closure, an update request may still exist even though no ontologically valid update pathway remains available. The present study refers to this condition as the *Limit of Ontological Updating*. The Limit of Ontological Updating does not imply that the capacity for updating has disappeared. Rather, it denotes a condition in which no update-capable world can be ontologically constituted within the subject's Mode of Existence.

4.4 Theoretical Correspondence with Depression

The present study does not attempt to redefine depression as a medical concept. From an ontological perspective, however, certain depressive states can be interpreted as manifestations of the Limit of Ontological Updating. In such cases, the subject has not necessarily lost meaning. Nor has the Future Fixed Point itself disappeared. Instead, the Mode of Existence that connects the subject to the Future Fixed Point has entered a state of local closure. Consequently, the object requiring updating is not the Future Fixed Point but the Mode of Existence itself.

4.5 An Ontological Interpretation of Terminal Choice

Terminal choice is not the consequence of the disappearance of the Future Fixed Point. Nor can it be explained solely through the loss of meaning. Instead, terminal choice may emerge as a local consequence when the Mode of Existence undergoes local closure to such an extent that no update-capable world can be constituted within the subject. This formulation enables a unified theoretical interpretation of terminal choice across fundamentally different existential structures, including military service, religious commitment, communal systems, prolonged social isolation, and depressive states.

4.6 The Role of the Premise Update Operating System

From the preceding discussion, it follows that the Premise Update Operating System is not a mechanism designed to prevent terminal choice directly. Nor is it a system that compels subjects to update themselves. Its fundamental role is to preserve, at the ontological level, the conditions under which the Mode of Existence continues to sustain the possibility of world generation rather than converging toward local closure. Within this framework, suffering is observed as an update request log. However, suffering itself does not guarantee that updating will occur. Updating becomes ontologically possible only when the Mode of Existence regains the capacity to constitute update-capable worlds.

4.7 Summary

This chapter has argued that terminal choice should not be regarded as a phenomenon produced by a single causal factor but as a local consequence derived from the Mode of Existence. Furthermore, certain depressive states can be interpreted as manifestations of the Limit of Ontological Updating, while the role of the Premise Update Operating System is not to command updating but to preserve the ontological foundation through which the Mode of Existence remains capable of generating worlds. Through this interpretation, the concepts of suffering as an update request introduced in FCT Paper No. 81, Pre-Judgment Determination introduced in FCT Paper No. 96, and the Mode of Existence introduced in the present study are integrated into a single ontological continuum that encompasses terminal choice.

5 The Meaning Density Tensor and the Implementation Specification of the Ontological Premise Update Operating System

Building upon the concepts of Ontological Determination, the Mode of Existence, and the Limit of Ontological Updating established in the preceding chapters, this chapter redefines the Meaning Density Tensor (MDT) as the implementation mechanism of the Ontological Premise Update Operating System. The objective of this chapter is not to interpret the Meaning Density Tensor as an evaluator of semantic quantity, but rather to establish it as the ontological information structure through which premise updating becomes implementable.

5.1 Redefinition of the Meaning Density Tensor

Previous formulations regarded the Meaning Density Tensor as an information structure for evaluating or representing the density of meaning maintained by a subject. From the ontological perspective developed in the present study, however, meaning itself is not a primary structure. Meaning emerges only after a world has been generated from the Mode of Existence and cognition subsequently becomes established within that world. Accordingly, the primary object represented by the Meaning Density Tensor is not meaning itself. The present study therefore defines the Meaning Density Tensor as follows.

The Meaning Density Tensor is an ontological information tensor that preserves, propagates, and updates the world-generative potential maintained by the Mode of Existence.

Under this definition, meaning density no longer denotes the quantity of meaning. Instead, it denotes the structural density through which the Mode of Existence preserves multiple possibilities for world generation while remaining ontologically open.

5.2 The Object of Ontological Implementation

The object implemented by the Meaning Density Tensor is neither knowledge, values, world-views, nor even the Future Fixed Point itself. Its implementation target is the world-generative

potential maintained by the Mode of Existence. Here, world-generative potential should not be understood as a collection of possible future choices. Ontologically, it refers to the condition in which the subject's world has not yet become closed into a single existential configuration. Accordingly, the Meaning Density Tensor is not a mechanism for enumerating future possibilities. Instead, it is an ontological implementation structure that guarantees that the Mode of Existence does not become fixed as a single closed world.

5.3 Correspondence with Suffering

FCT Paper No. 81 defined suffering as an update request log. The present study preserves this definition without modification. When interpreted through the Meaning Density Tensor, however, suffering should not be understood as a decrease in meaning. Nor should it be interpreted as a reduction in happiness or subjective well-being. Rather, suffering is an ontological system log indicating that the world-generative potential maintained by the Mode of Existence is locally converging toward closure. Accordingly, suffering is not an abnormal condition. It is informational evidence that the world-generative potential represented by the Meaning Density Tensor is undergoing local contraction.

5.4 Correspondence with Premise Updating

If the Meaning Density Tensor is an ontological information structure, premise updating cannot be interpreted as rewriting meaning. Nor is it merely a modification of cognitive structures. Premise updating is an ontological operation that reconstructs the world-generative potential maintained by the Mode of Existence. As a consequence of this reconstruction, meaning, value, the Future Fixed Point, self-image, judgment, cognition, and action are simultaneously reorganized. Accordingly, the Premise Update Operating System should not be understood as rewriting the Meaning Density Tensor itself. Instead, it reconstructs the world-generative potential represented through the Meaning Density Tensor.

5.5 Design Requirements for the Ontological Premise Update Operating System

The preceding discussion implies that an Ontological Premise Update Operating System must satisfy the following design requirements.

1. It must not reduce the Mode of Existence to cognitive structures.
2. It must interpret suffering as local contraction of world-generative potential rather than as an abnormal signal.
3. It must not compel subjects to update themselves.
4. It must not assume updating capability as a prerequisite.
5. It must implement the world-generative potential maintained by the Mode of Existence as an information structure capable of continuous preservation.

6. It must treat the Meaning Density Tensor as a structure for preserving world-generative potential rather than semantic quantity.
7. It must treat the Future Fixed Point not as a fixed objective but as a convergence structure derived from the Mode of Existence.

5.6 Implementation Specification

Based on the theoretical framework developed thus far, the implementation specification of the Ontological Premise Update Operating System can be summarized as follows.

1. Ontological Determination gives rise to the Mode of Existence.
2. The Mode of Existence preserves world-generative potential.
3. The Meaning Density Tensor represents this world-generative potential as an ontological information structure.
4. The World OS implements the world through the Meaning Density Tensor.
5. The Personal OS derives cognition, judgment, and action from the World OS.
6. Suffering is observed as a local contraction of world-generative potential represented within the Meaning Density Tensor.
7. Premise updating is realized through the reconstruction of the world-generative potential maintained by the Mode of Existence.

Within this specification, updating is not an externally imposed command. Rather, updating emerges naturally as an ontological phenomenon once the Mode of Existence regains the capacity to sustain world-generative potential.

5.7 Summary

This chapter has redefined the Meaning Density Tensor not as an evaluator of semantic information but as the ontological information structure that implements the world-generative potential maintained by the Mode of Existence. Furthermore, the Ontological Premise Update Operating System has been characterized not as a mechanism that instructs subjects to update themselves, but as an operating system that preserves, propagates, and reconstructs world-generative potential. Through this reformulation, the Premise Update Operating System introduced in FCT Paper No. 81, the concept of Pre-Judgment Determination established in FCT Paper No. 96, and the Mode of Existence proposed in the present study are integrated into a unified ontological implementation framework mediated by the Meaning Density Tensor.

6 Design Principles of the Ontological Premise Update Operating System

This chapter integrates the concepts of Ontological Determination, the Mode of Existence, and the Meaning Density Tensor developed in the preceding chapters in order to formulate the design principles of the Ontological Premise Update Operating System (Ontological Premise Update OS). The objective of this chapter is to specify precisely what the operating system is intended to implement while rigorously distinguishing ontological implementation from informational implementation.

6.1 Hierarchical Separation of the Design Targets

The present study does not treat the implementation target of the operating system as a single homogeneous layer. Ontology and information structures perform fundamentally different functions and therefore must be distinguished. Accordingly, the implementation target is divided into the following four layers:

$$\mathcal{O} \rightarrow \mathcal{M} \rightarrow \mathcal{I} \rightarrow \mathcal{S}$$

where

$\mathcal{O}$ Ontological Layer

$\mathcal{M}$ Mode-of-Existence Layer

$\mathcal{I}$ Information Implementation Layer

$\mathcal{S}$ System Implementation Layer

Failure to distinguish these layers inevitably reduces ontology to information processing, thereby contradicting the ontological position adopted throughout Future Convergence Theory.

6.2 The Ontological Layer

The Ontological Layer is neither implemented nor updated. At this level, only the axiom of Pre-Judgment Determination is assumed. Existence is already in a state of determination prior to judgment. This principle is not intended to be demonstrated or derived. Rather, it functions as the foundational condition upon which the entire theoretical framework depends. Accordingly, concepts such as meaning, value, cognition, the future, the self, society, and culture do not exist at this layer. All of these emerge only within lower layers of the hierarchy.

6.3 The Mode-of-Existence Layer

The Mode-of-Existence Layer represents the temporal unfolding of the Ontological Layer. The Mode of Existence is neither a belief held by the subject nor a system of values. Instead, it is the world within which the subject already exists. Consequently, world premises do not exist independently of the Mode of Existence. The Mode of Existence itself constitutes the subject's

world premises. Premise updating is therefore equivalent to updating the Mode of Existence, and updating the Mode of Existence is, in turn, equivalent to updating the world premises.

6.4 The Information Implementation Layer

The Information Implementation Layer represents the Mode of Existence as information. Only at this stage is the Meaning Density Tensor introduced. The Meaning Density Tensor is not identical to the Mode of Existence. Rather, it functions as the informational medium through which the Mode of Existence is represented while propagating the structure of world generation. Accordingly, modifying the Meaning Density Tensor does not directly modify the Mode of Existence itself. Instead, the Meaning Density Tensor serves as the informational substrate through which updates of the Mode of Existence become implementable. This distinction constitutes one of the central design principles proposed in the present study.

6.5 The System Implementation Layer

Within the System Implementation Layer, the Meaning Density Tensor is employed to implement the World OS, the Personal OS, the Future Fixed Point, cognition, judgment, and action. Only at this layer do concepts such as updating, suffering, convergence toward the Future Fixed Point, evaluation, responsibility, meaning, institutions, and society become operational. Consequently, the operating system does not manipulate ontology directly. Nor does it directly manipulate the Mode of Existence. Instead, its direct object of operation is the informational implementation represented through the Meaning Density Tensor.

6.6 Ontological Updating and Informational Updating

The present study rigorously distinguishes Ontological Updating from Informational Updating. Informational Updating refers to structural modifications within the Meaning Density Tensor. Ontological Updating, by contrast, denotes the updating of the Mode of Existence itself. Informational Updating may represent Ontological Updating, but it is not identical to it. Conversely, whenever Ontological Updating occurs, the entire informational implementation is necessarily reconstructed. Accordingly, Informational Updating constitutes a necessary condition for expressing Ontological Updating, but it is not a sufficient condition.

6.7 The Design Specification of Suffering

Suffering is not an abnormal signal generated by the operating system. Instead, it is an ontological consistency log indicating that the discrepancy between the world-generative structure maintained by the Mode of Existence and its current informational implementation has exceeded a certain threshold. Accordingly, suffering should not be regarded as an object to be eliminated. Rather, it is an observable indication that the informational structure is progressively losing ontological consistency with the Mode of Existence.

6.8 Fundamental Design Axioms

From the foregoing discussion, the Ontological Premise Update Operating System must satisfy the following fundamental design axioms.

1. The Ontological Layer must never be reduced to information processing.
2. The Mode of Existence must never be reduced to cognitive structures.
3. The Meaning Density Tensor is not the Mode of Existence itself but its informational implementation.
4. The operating system directly manipulates only the Meaning Density Tensor.
5. Suffering is treated as an ontological consistency log.
6. Updating is implemented as the restoration of ontological consistency with the Mode of Existence.
7. The Future Fixed Point is not a target generated by the operating system but a convergence structure derived from the Mode of Existence.

6.9 Definition of the Ontological Premise Update Operating System

The present study defines the Ontological Premise Update Operating System as follows.

The Ontological Premise Update Operating System is an operating-system framework that implements the Mode of Existence derived from Ontological Determination through the Meaning Density Tensor as an informational structure while continuously maintaining ontological consistency. Within this framework, premise updating does not refer to local modifications of the Meaning Density Tensor itself. Rather, it denotes the phenomenon in which the entire informational structure is reimplemented as a consequence of reconstructing its consistency with the Mode of Existence.

This definition unifies the Premise Update Operating System introduced in FCT Paper No. 81, the concept of Pre-Judgment Determination established in FCT Paper No. 96, and the concepts of the Mode of Existence, the Meaning Density Tensor, and the Limit of Ontological Updating developed in the present study into a single ontological operating-system framework.

6.10 Summary

This chapter distinguished ontology, the Mode of Existence, informational implementation, and system implementation as four independent layers, thereby clarifying the respective roles of ontological implementation and informational implementation. Furthermore, it demonstrated that the Meaning Density Tensor should not be regarded as the Mode of Existence itself but rather as the informational medium through which the Mode of Existence is implemented. Consequently, the Premise Update Operating System within Future Convergence Theory is redefined not merely as an updating algorithm but as an implementation framework for maintaining ontological consistency.

7 Ontological Judgment and the Mode of Existence

This chapter adopts the axiom of *pre-judgment ontological determination*, introduced in FCT Paper 96, as the ontological foundation of the present study and clarifies its relationship with the concept of the *mode of existence*. The objective of this chapter is to redefine the object of premise updating, not as a cognitive structure, but as an ontological structure.

7.1 The Ontological Status of Pre-Judgment Determination

Conventional cognitive science generally assumes that perception and cognition proceed according to the causal sequence

$$\text{World} \rightarrow \text{Perception} \rightarrow \text{Cognition} \rightarrow \text{Judgment}.$$

Within this framework, the world exists independently of the subject, while cognition functions as a process that internally reconstructs external information.

FCT rejects this ordering.

Instead, FCT assumes that existence is not fundamentally a cognitive agent that subsequently performs judgments. Rather, existence is already constituted as an ontologically determined state prior to any cognitive operation.

Accordingly, the ontological order becomes

$$\text{Existence} \rightarrow \text{Ontological Determination} \rightarrow \text{World} \rightarrow \text{Cognition} \rightarrow \text{Judgment}.$$

The term *determination* here does not refer to logical reasoning, emotional evaluation, accumulated experience, acquired knowledge, or learned behavior.

Instead, it denotes the ontological condition under which existence itself is already constituted before cognition becomes possible.

Consequently, pre-judgment determination is not introduced as an empirical hypothesis requiring proof but as an ontological axiom occupying the same foundational level as existence itself.

7.2 Introducing the Mode of Existence

Although pre-judgment determination establishes the ontological basis of existence, it does not by itself describe the continuity of existence through time.

To bridge this gap, the present study introduces the concept of the *mode of existence*.

The mode of existence is defined as the present state produced through the temporal unfolding of ontological determination.

Importantly, the mode of existence is not generated through cognition.

Rather, cognition, values, meaning, future images, self-images, social roles, and all higher-order cognitive structures are derived from the mode of existence.

Therefore, the mode of existence occupies a more fundamental ontological level than cognition.

7.3 The Mode of Existence and World Premises

In conventional discourse, a worldview is generally regarded as a system of beliefs or values adopted by an individual.

From the ontological perspective proposed here, however, a subject does not merely adopt a worldview.

Rather, the subject already exists as that world.

Accordingly, world premises are not cognitive objects selected through reasoning.

They are identical with the subject's mode of existence itself.

Religious systems, national identity, social institutions, communities, self-realization, and interpersonal evaluation are therefore understood as concrete worlds generated from different modes of existence.

Changing a world premise consequently does not mean modifying a belief system.

Instead, it requires an ontological update of the underlying mode of existence from which that world is generated.

7.4 The Mode of Existence and the Formation Process

The mode of existence is not produced simply through the accumulation of experiences over the course of one's life.

Nor is it determined solely by cognitive learning.

Nevertheless, it possesses temporal continuity.

The present study interprets the mode of existence as the present manifestation of ontological determination unfolding through time.

From the perspective of the subject, this temporal unfolding is experienced as what is ordinarily called one's life history or formation process.

Accordingly, the formation process is not regarded as the cause of the mode of existence.

Rather, it is the observable manifestation of the temporal development of the mode of existence itself.

Thus, instead of assuming

Mode of Existence $\rightarrow$ Formation Process,

this study interprets

Ontological Determination = Mode of Existence = Formation Process

as three different observational descriptions of the same ontological structure.

7.5 Updating the Mode of Existence

Premise updating has traditionally been understood as an operation that modifies meaning structures or cognitive structures.

Once the concept of the mode of existence is introduced, however, cognition is no longer the primary object of updating.

Instead, the object of updating becomes the mode of existence itself.

When the mode of existence changes, all higher-order structures derived from it—including meaning, values, purposes, future fixed points, judgments, and behavior—are reconstructed simultaneously.

Premise updating is therefore understood not as a sequential correction of individual cognitive elements but as an ontological reconstruction of the foundational condition from which cognition itself emerges.

7.6 Ontological Updating versus Cognitive Updating

Cognitive updating modifies individual knowledge, evaluations, and interpretations while leaving the surrounding world unchanged.

Formally,

$$\text{World} = \text{Constant},$$

while cognition alone is altered.

Ontological updating follows a fundamentally different structure:

$$\text{Mode of Existence} \implies \text{World} \implies \text{Cognition} \implies \text{Judgment}.$$

Here, updating the mode of existence simultaneously transforms the generated world, cognitive structures, and subsequent judgments.

Therefore, ontological updating necessarily contains cognitive updating as one of its consequences.

The converse, however, does not hold.

Cognitive updating alone cannot reconstruct the underlying mode of existence.

7.7 Definition of Ontological Updating

Based on the preceding discussion, this study defines *ontological updating* as follows.

Definition 1 (Ontological Updating). *Ontological updating is the ontological operation in which the mode of existence itself is reconstructed, thereby simultaneously reorganizing all structures derived from it, including world premises, meaning, future fixed points, values, cognition, judgment, and behavior.*

Under this definition, the Premise Update Operating System introduced in FCT Paper 81 is no longer interpreted as a cognitive updating mechanism.

Instead, it is redefined as an operating system that updates the mode of existence itself.

Furthermore, the pre-judgment determination introduced in FCT Paper 96 becomes connected to the Premise Update Operating System through the mediating concept of the mode of existence.

As a result, ontology, premise updating, meaning density, future fixed points, and the World OS are unified into a single ontological continuum.

8 Terminal Choice, Depression, and the Limits of Ontological Updating

This chapter examines depression and terminal choice from the perspective of the Ontological Premise Update Operating System developed in the preceding chapters. The purpose of this chapter is neither to redefine depression as a medical condition nor to explain terminal choice through a single causal factor. Instead, the objective is to clarify the ontological structures through which terminal choice may emerge from the relationship between the mode of existence and the Premise Update Operating System.

8.1 Terminal Choice Is Not an Ontological Axiom

Terminal choice is often explained in terms of factors such as depression, loss of meaning, social isolation, external evaluation, or self-negation.

From the ontological perspective adopted in this study, however, none of these factors constitutes either a necessary or a sufficient condition.

For example, soldiers whose highest commitment is national service, individuals who willingly accept self-sacrifice based on religious convictions, subjects embedded within strong communal norms, and individuals experiencing prolonged social isolation may possess fundamentally different modes of existence. Nevertheless, under certain conditions, each may arrive at terminal choice.

Conversely, many individuals exposed to similar external circumstances never reach such a conclusion.

Accordingly, terminal choice cannot be understood as the direct consequence of any single psychological, social, or environmental factor.

Instead, it is interpreted as a local consequence derived from a particular mode of existence.

For this reason, the present study does not treat terminal choice itself as an ontological axiom.

8.2 The Mode of Existence and World Closure

The mode of existence constitutes the world that is ontologically established for a subject.

Consequently, world premises are not beliefs voluntarily adopted by the subject but are identical with the subject's own mode of existence.

As long as the mode of existence preserves the possibility of updating, the world remains ontologically open.

However, when the mode of existence converges toward local closure, the subject experiences the existing world as the only possible reality.

The notion of closure employed here does not denote logical uniqueness.

Rather, it refers to the condition in which alternative modes of world generation cease to exist ontologically for the subject.

8.3 The Limits of Ontological Updating

FCT Paper 81 defined suffering as an update request indicating divergence from a future fixed point.

The present study preserves this interpretation.

Nevertheless, the existence of an update request does not logically imply the existence of an available update pathway.

Human beings are influenced by numerous factors, including neurophysiological conditions, cognitive resources, social environments, and long-term experiential histories.

As these factors interact, the mode of existence may gradually converge toward local closure.

In such circumstances, update requests continue to exist, while ontologically accessible update paths no longer emerge.

The present study defines this condition as the *limit of ontological updating*.

The limit of ontological updating does not imply that the capacity for updating has disappeared.

Rather, it denotes a condition in which no ontologically realizable world permitting further updating is generated.

8.4 Theoretical Correspondence with Depression

This study does not attempt to redefine depression as a medical diagnosis.

Nevertheless, from the present ontological perspective, certain depressive states may be interpreted as manifestations of the limit of ontological updating.

Importantly, the subject has not necessarily lost meaning itself.

Nor has the future fixed point ceased to exist ontologically.

Instead, the mode of existence connecting the subject to that future fixed point has undergone local closure.

Consequently, the proper object of updating is not the future fixed point itself but the underlying mode of existence.

8.5 An Ontological Interpretation of Terminal Choice

Terminal choice is not understood as the consequence of the disappearance of the future fixed point.

Nor can it be sufficiently explained by the loss of meaning alone.

Rather, terminal choice may emerge when the mode of existence locally closes and no updateable world remains ontologically realizable for the subject.

This interpretation provides a unified theoretical framework capable of describing terminal choice across heterogeneous contexts, including military duty, religious commitment, communal obligations, prolonged social isolation, and depressive conditions.

8.6 The Role of the Premise Update Operating System

Within this framework, the Premise Update Operating System is not designed to prevent terminal choice directly.

Neither is it intended to compel subjects to update themselves.

Its role is to maintain the ontological conditions under which the mode of existence continues to preserve the possibility of world generation rather than converging toward irreversible local closure.

In this context, suffering functions as an observable update request log.

However, suffering alone cannot guarantee that updating will occur.

Updating becomes ontologically possible only when the mode of existence once again generates an updateable world.

8.7 Summary

This chapter has argued that terminal choice should not be regarded as the inevitable consequence of any single causal factor but rather as a local consequence derived from a particular mode of existence.

Furthermore, certain depressive states have been interpreted as manifestations of the limit of ontological updating.

Accordingly, the function of the Premise Update Operating System is not to command updating but to preserve the ontological foundation that maintains the possibility of world generation.

Through this reformulation, the update-request interpretation of suffering proposed in FCT Paper 81, the axiom of pre-judgment determination established in FCT Paper 96, and the concept of the mode of existence introduced in the present study are integrated into a single ontological continuum.

9 The Meaning Density Tensor and the Implementation Architecture of the Ontological Premise Update Operating System

This chapter redefines the Meaning Density Tensor (MDT) as the implementation architecture of the Ontological Premise Update Operating System by integrating the concepts developed in the previous chapters, namely ontological determination, the mode of existence, and the limit of ontological updating.

The objective of this chapter is not to regard the Meaning Density Tensor as a metric for evaluating semantic quantity, but rather to establish it as the informational structure through which ontological premise updating becomes implementable.

9.1 Redefinition of the Meaning Density Tensor

Previous studies have generally interpreted the Meaning Density Tensor as an informational structure representing the density or quantity of meaning maintained by an individual.

From the ontological perspective adopted in this study, however, meaning itself is not a primary structure.

Meaning emerges only after the mode of existence generates a world in which cognition subsequently becomes possible.

Accordingly, the Meaning Density Tensor does not preserve meaning itself.

Instead, this study defines it as follows.

Definition 2 (Meaning Density Tensor). *The Meaning Density Tensor is an ontological information tensor that preserves, propagates, and updates the world-generating potential maintained by the mode of existence.*

Under this definition, meaning density no longer refers to the amount of semantic information possessed by a subject.

Instead, it represents the degree to which the mode of existence preserves multiple ontological possibilities for world generation.

9.2 The Ontological Object of Implementation

The implementation target of the Meaning Density Tensor is neither knowledge, values, world-views, nor the future fixed point itself.

Its implementation target is the world-generating potential maintained by the mode of existence.

Here, world-generating potential does not refer to a collection of future alternatives or decision branches.

Ontologically, it denotes the condition in which the subject's world has not become completely closed.

Accordingly, the Meaning Density Tensor is not an engine for enumerating future possibilities.

Rather, it functions as the informational structure that preserves the ontological openness required for multiple worlds to remain generable.

9.3 Correspondence with Suffering

FCT Paper 81 defined suffering as an update request log.

The present study preserves this interpretation.

However, from the standpoint of the Meaning Density Tensor, suffering should not be understood as a reduction in semantic quantity.

Nor should it be interpreted as a decrease in subjective happiness.

Instead, suffering represents an ontological system log indicating that the world-generating potential maintained by the mode of existence is locally converging toward closure.

Consequently, suffering is not an abnormality.

Rather, it is informational evidence that the world-generating potential represented by the Meaning Density Tensor is undergoing local contraction.

9.4 Correspondence with Premise Updating

If the Meaning Density Tensor is understood as an ontological information structure, premise updating cannot be interpreted as rewriting meaning.

Nor is it merely a modification of cognitive structures.

Premise updating is the ontological reconstruction of the world-generating potential maintained by the mode of existence.

As a consequence of this reconstruction, meaning, values, future fixed points, self-image, judgment, cognition, and behavior are reorganized simultaneously.

Accordingly, the Premise Update Operating System does not directly rewrite the Meaning Density Tensor itself.

Instead, it reconstructs the world-generating potential represented through the Meaning Density Tensor.

9.5 Design Requirements for the Ontological Premise Update Operating System

The preceding discussion implies that the Ontological Premise Update Operating System must satisfy the following design requirements.

1. The mode of existence must never be reduced to cognitive structures.
2. Suffering must be interpreted as a local contraction of world-generating potential rather than as an abnormal signal.
3. Updating must not be imposed upon the subject.
4. Updating capability must not be assumed as a prerequisite.
5. The world-generating potential maintained by the mode of existence must be implemented as a continuously sustainable informational structure.
6. The Meaning Density Tensor must represent world-generating potential rather than semantic quantity.
7. The future fixed point must be treated as a convergence structure derived from the mode of existence rather than as a predetermined objective.

9.6 Implementation Architecture

The fundamental implementation architecture of the Ontological Premise Update Operating System can therefore be summarized as follows.

1. Ontological determination gives rise to the mode of existence.
2. The mode of existence preserves world-generating potential.

3. The Meaning Density Tensor maintains this world-generating potential as an informational structure.
4. The World OS implements the experienced world through the Meaning Density Tensor.
5. The Personal OS derives cognition, judgment, and behavior from the World OS.
6. Suffering is observed as the local contraction of world-generating potential represented within the Meaning Density Tensor.
7. Premise updating is realized through the reconstruction of the world-generating potential maintained by the mode of existence.

Within this implementation architecture, updating is not an externally imposed command.

Rather, updating emerges naturally as an ontological phenomenon whenever the mode of existence once again preserves the possibility of world generation.

9.7 Summary

This chapter has redefined the Meaning Density Tensor as an ontological information structure implementing the world-generating potential maintained by the mode of existence, rather than as a measure of semantic information.

Furthermore, the Ontological Premise Update Operating System has been characterized not as an operating system that compels updating, but as one that preserves, propagates, and reconstructs world-generating potential.

Through this reformulation, the Premise Update Operating System introduced in FCT Paper 81, the principle of pre-judgment determination established in FCT Paper 96, and the concept of the mode of existence proposed in the present study are integrated into a unified ontological implementation framework mediated by the Meaning Density Tensor.

10 Design Principles of the Ontological Premise Update Operating System

This chapter integrates the concepts developed in the preceding chapters—ontological determination, the mode of existence, and the Meaning Density Tensor—to establish the design principles of the Ontological Premise Update Operating System (OPUOS).

The objective of this chapter is to specify the proper object of implementation while rigorously distinguishing ontological implementation from informational implementation.

10.1 Hierarchical Separation of Design Objects

This study does not treat the implementation target of the operating system as belonging to a single layer.

Ontology and information structures perform fundamentally different functions and therefore must be distinguished.

Accordingly, the implementation hierarchy is divided into the following four layers:

$$\mathcal{O} \rightarrow \mathcal{M} \rightarrow \mathcal{I} \rightarrow \mathcal{S},$$

where

$\mathcal{O}$ Ontological Layer

$\mathcal{M}$ Mode of Existence Layer

$\mathcal{I}$ Information Implementation Layer

$\mathcal{S}$ System Implementation Layer

Failure to distinguish these layers would reduce ontology itself to information processing, thereby contradicting the ontological position adopted by Future Convergence Theory.

10.2 The Ontological Layer

The ontological layer is neither implemented nor updated.

At this level, the only adopted axiom is pre-judgment ontological determination.

Existence is already constituted as a state of ontological determination prior to any cognitive activity.

This principle is not a theorem requiring proof but the foundational condition upon which the entire theoretical framework rests.

Consequently, concepts such as meaning, value, cognition, future, self, society, and culture do not belong to this layer.

They emerge only within lower layers of the hierarchy.

10.3 The Mode of Existence Layer

The mode of existence layer represents the temporal unfolding of the ontological layer.

The mode of existence is neither a belief system nor a value system maintained by the subject.

Rather, it is the world within which the subject already exists.

Accordingly, world premises do not exist independently of the mode of existence.

Instead, the mode of existence itself constitutes the subject's world premises.

Premise updating is therefore equivalent to updating the mode of existence, and updating the mode of existence is equivalent to updating the subject's world premises.

10.4 The Information Implementation Layer

The information implementation layer provides an informational representation of the mode of existence.

It is at this layer that the Meaning Density Tensor is introduced.

The Meaning Density Tensor is not identical with the mode of existence itself.

Rather, it serves as the informational medium through which the mode of existence is represented and through which world-generating structures are propagated.

Consequently, modifying the Meaning Density Tensor does not directly alter the mode of existence.

Instead, the Meaning Density Tensor functions as the informational substrate that enables the implementation of ontological updating.

This distinction constitutes one of the central design principles of the present theory.

10.5 The System Implementation Layer

Within the system implementation layer, the Meaning Density Tensor provides the informational basis for implementing the World OS, the Personal OS, future fixed points, cognition, judgment, and behavior.

Only at this stage do concepts such as updating, suffering, future convergence, evaluation, responsibility, meaning, institutions, and society emerge.

Accordingly, the operating system neither manipulates ontology directly nor updates the mode of existence itself.

Its direct object of operation is exclusively the informational implementation represented by the Meaning Density Tensor.

10.6 Ontological Updating and Informational Updating

This study draws a strict distinction between ontological updating and informational updating.

Informational updating refers to structural modifications within the Meaning Density Tensor.

Ontological updating, by contrast, denotes the reconstruction of the mode of existence itself.

Informational updating can represent ontological updating, but it is not identical with ontological updating.

Conversely, whenever ontological updating occurs, the entire informational implementation is necessarily reconstructed.

Therefore, informational updating constitutes a necessary condition for implementation but is not a sufficient condition for ontological updating.

10.7 The Design Specification of Suffering

Suffering is not an abnormal signal generated by the operating system.

Rather, suffering is an ontological consistency log indicating that the discrepancy between the world-generating structure maintained by the mode of existence and its informational implementation has exceeded a critical threshold.

Accordingly, suffering is not an object to be eliminated.

Instead, it represents the observable consequence of decreasing ontological consistency between the mode of existence and the implemented informational structure.

10.8 Fundamental Design Axioms

Based on the preceding analysis, the Ontological Premise Update Operating System must satisfy the following fundamental axioms.

1. The ontological layer shall never be reduced to information processing.
2. The mode of existence shall never be reduced to cognitive structures.
3. The Meaning Density Tensor represents the informational implementation of the mode of existence rather than the mode of existence itself.
4. The operating system directly manipulates only the Meaning Density Tensor.
5. Suffering shall be interpreted as an ontological consistency log.
6. Updating shall be implemented as the restoration of consistency between the informational implementation and the mode of existence.
7. The future fixed point shall be understood as a convergence structure derived from the mode of existence rather than as an externally imposed objective.

10.9 Definition of the Ontological Premise Update Operating System

The present study defines the Ontological Premise Update Operating System as follows.

Definition 3 (Ontological Premise Update Operating System). *The Ontological Premise Update Operating System is an operating system that implements the mode of existence derived from ontological determination through the Meaning Density Tensor as an informational structure while continuously maintaining ontological consistency.*

Premise updating does not denote a local modification of the Meaning Density Tensor itself. Rather, it denotes the phenomenon in which the entire informational structure is reimplemented as ontological consistency with the mode of existence is reconstructed.

Under this definition, the Premise Update Operating System introduced in FCT Paper 81, the principle of pre-judgment determination introduced in FCT Paper 96, and the concepts of the mode of existence, the Meaning Density Tensor, and the limit of ontological updating introduced in the present study are unified into a single ontological operating system.

10.10 Summary

This chapter established a four-layer architecture separating ontology, the mode of existence, informational implementation, and system implementation, thereby clarifying the distinction between ontological implementation and informational implementation.

Furthermore, it demonstrated that the Meaning Density Tensor is not identical with the mode of existence but instead serves as the informational medium through which the mode of existence is implemented.

Consequently, the Premise Update Operating System in Future Convergence Theory is redefined not as a conventional updating algorithm but as an implementation architecture dedicated to maintaining ontological consistency.

11 Discussion

This chapter discusses the theoretical implications of the Ontological Premise Update Operating System developed in the preceding chapters, with particular attention to the relationships among the mode of existence, the Meaning Density Tensor, premise updating, suffering, depression, and terminal choice.

The objective of this discussion is neither to replace existing psychological theories nor to redefine psychiatric diagnoses. Rather, it is to clarify, from an ontological perspective, why premise updating is required, why the Meaning Density Tensor is necessary as an implementation medium, and why the preservation of world-generating potential constitutes a fundamental design requirement for the operating system proposed in this study.

11.1 The Fundamental Problem Addressed by the Ontological Premise Update Operating System

Conventional models of updating implicitly assume that subjects remain capable of updating themselves.

That is, they presuppose that individuals possess sufficient cognitive resources to reconstruct meaning and establish new future fixed points.

However, when depression and terminal choice are reconsidered within an ontological framework, the central issue is no longer updating capability itself.

Even when an individual retains cognitive capacity, updating cannot occur if the mode of existence has converged toward local closure.

Conversely, even under conditions of limited cognitive capability, updating may remain ontologically possible if the mode of existence continues to preserve world-generating potential.

Accordingly, the proper object of operating-system design is not updating capability itself but the preservation of world-generating potential maintained by the mode of existence.

11.2 Ontological Closure and Terminal Choice

The present study interprets terminal choice as one possible consequence of local closure within the mode of existence.

Importantly, local closure does not imply absolute closure.

Objectively, alternative possibilities may continue to exist.

Nevertheless, if the subject's mode of existence no longer generates those possibilities as ontologically realizable worlds, then the experienced world has effectively become closed.

This distinction allows terminal choice to be interpreted not as the disappearance of objective possibilities but as the disappearance of ontological possibilities available to the subject.

Consequently, situations in which external alternatives objectively existed but could not be experienced as genuine possibilities by the subject become theoretically explainable within a unified ontological framework.

11.3 The Ontological Status of Others, Society, and Religion

Within the present framework, other persons, society, nations, religions, and communities are not regarded as direct causes of terminal choice.

Nor are they considered components that constitute the mode of existence itself.

Instead, they are interpreted as elements of the world generated from a particular mode of existence.

Accordingly, religious self-sacrifice, national loyalty, communal obligations, social evaluation, and self-realization are understood as world premises derived from different modes of existence.

None of these therefore qualifies as an ontological axiom.

The only ontological axiom adopted in the present theory remains pre-judgment ontological determination.

11.4 Comparison with Prolonged Social Isolation

Prolonged social isolation serves in this study as a counterexample for examining the relationship between the mode of existence and terminal choice.

Individuals experiencing prolonged isolation may exhibit substantially reduced interaction with society, diminished social evaluation, fewer competitive pressures, and weakened institutional roles.

Nevertheless, such conditions alone neither necessarily suppress nor inevitably promote terminal choice.

From the ontological perspective, the decisive factor is not the degree of social connection itself but whether the mode of existence continues to preserve world-generating potential.

Therefore, the essential question concerns not social isolation per se but the ontological structure through which the subject's world continues to exist.

11.5 The Ontological Necessity of the Meaning Density Tensor

Within the present framework, the Meaning Density Tensor is neither a measure of semantic quantity nor the mode of existence itself.

Its necessity arises because ontology cannot directly function as information processing, whereas an operating system must necessarily be implemented as an informational structure.

Accordingly, an intermediate informational layer must connect ontology with system implementation.

The Meaning Density Tensor fulfills precisely this role.

Through this intermediate layer, ontological updating and informational updating become theoretically distinguishable while remaining systematically connected.

11.6 The Function of the Premise Update Operating System

The preceding discussion clarifies that the Premise Update Operating System is not designed to force subjects to update themselves.

Neither does it function by providing externally imposed meaning.

Instead, its role is to implement and preserve, as an informational structure, the world-generating potential maintained by the mode of existence while sustaining ontological consistency.

Within this framework, suffering functions as a consistency log indicating local contraction of world-generating potential, whereas updating is understood as the phenomenon in which consistency between the mode of existence and its informational implementation is restored.

11.7 Theoretical Implications

The present study extends the Premise Update Operating System beyond conventional theories of cognitive updating into a theory of ontological implementation.

Furthermore, the Meaning Density Tensor is redefined not as a semantic evaluation structure but as the ontological implementation medium connecting the mode of existence with informational structures.

Consequently, pre-judgment ontological determination, the mode of existence, world premises, the Meaning Density Tensor, the World OS, premise updating, suffering, and future fixed points should no longer be regarded as independent concepts.

Instead, they collectively constitute a unified ontological implementation architecture.

This integration clarifies the theoretical relationship between ontology and operating-system theory within Future Convergence Theory while providing a systematic foundation for future theoretical development and implementation design.

12 Conclusion

This study has reconstructed the Premise Update Operating System (PUOS) within Future Convergence Theory (FCT) as an ontological implementation architecture rather than as a model of cognitive updating.

To accomplish this, the study adopted the Premise Update Operating System proposed in FCT Paper 81 and the principle of pre-judgment ontological determination established in FCT Paper 96 as its theoretical foundations. Building upon these foundations, the concepts of the mode of existence, the Meaning Density Tensor, the limit of ontological updating, and terminal choice were integrated into a unified ontological framework.

The principal findings of the present study are summarized below.

12.1 Clarification of the Object of Ontological Updating

Premise updating has conventionally been interpreted as an operation that modifies cognition, meaning, evaluation, or representations of the future.

Once the ontological hierarchy proposed in this study is introduced, however, cognition is no longer the primary object of updating.

Instead, the object of updating is the mode of existence itself.

Meaning, value, future fixed points, cognition, judgment, and behavior are not independent entities but higher-order structures derived from the mode of existence.

Accordingly, they are reconstructed simultaneously whenever the mode of existence is updated.

12.2 The Ontological Position of the Meaning Density Tensor

The present study does not define the Meaning Density Tensor as a structure for evaluating semantic information.

Instead, it is positioned as the ontological implementation medium connecting the mode of existence with informational structures.

Ontology itself does not perform information processing.

An operating system, however, necessarily operates as an informational system.

The Meaning Density Tensor therefore serves as the informational interface that bridges these two fundamentally different domains.

12.3 Redefinition of Suffering

Suffering is not interpreted as an abnormal signal.

Instead, it functions as an ontological consistency log indicating that consistency between the mode of existence and its informational implementation has become locally disrupted.

Accordingly, suffering itself is not an object to be eliminated.

Rather, it represents an observable indication that ontological updating has become necessary.

12.4 The Position of Terminal Choice

Terminal choice cannot be adequately explained by any single factor, including loss of meaning, social isolation, religion, national commitment, external evaluation, or self-realization.

Instead, this study positions terminal choice as a local consequence that may emerge from local closure within the mode of existence.

This interpretation makes it possible to describe diverse social structures, cultural systems, value systems, and belief systems within a single ontological framework.

12.5 Ontological Updatability

Perhaps the most significant contribution of the present study is the integration of the concepts of world-generating potential, ontological openness, and premise updatability into a more fundamental ontological concept.

To achieve this integration, the study introduces the concept of *Ontological Updatability*.

Ontological Updatability refers to the ontological property through which existence avoids fixation into a single closed structure while preserving the possibility of continuous updating under conditions of ontological consistency.

Importantly, Ontological Updatability should not be confused with updating capability.

Nor is it equivalent to cognitive ability, accumulated knowledge, or learning capacity.

Instead, Ontological Updatability is proposed as an intrinsic ontological property of existence itself.

12.6 Final Definition of the Ontological Premise Update Operating System

Based on the theoretical developments presented throughout this study, the Ontological Premise Update Operating System is defined as follows.

Definition 4 (Ontological Premise Update Operating System). *The Ontological Premise Update Operating System is an operating system that implements the mode of existence established through pre-judgment ontological determination as an informational structure by means of the Meaning Density Tensor while continuously preserving Ontological Updatability.*

Suffering functions as an observational log indicating local disruption of ontological consistency.

Premise updating denotes the phenomenon in which the entire informational structure is reimplemented as Ontological Updatability becomes re-established.

12.7 Future Work

The present study introduces Ontological Updatability as a fundamental ontological property.

Its rigorous mathematical formulation, however, remains an open problem.

Future research should investigate how Ontological Updatability may be represented within the Meaning Density Tensor through information-theoretic measures, topological structures, tensor fields, or convergence conditions.

Furthermore, although the mode of existence has been formulated here as an individual ontological concept, future work should extend the theory to interactions among multiple subjects as well as to organizational operating systems, social operating systems, and civilizational operating systems.

12.8 Concluding Remarks

This study integrates the Premise Update Operating System introduced in FCT Paper 81 with the principle of pre-judgment ontological determination established in FCT Paper 96 through the newly proposed concept of Ontological Updatability.

As a consequence, the Premise Update Operating System within Future Convergence Theory is no longer understood as a cognitive updating mechanism.

Instead, it is redefined as an ontological operating system that implements, preserves, and maintains the updateability inherent in existence itself through informational structures.

Accordingly, ontology, the Meaning Density Tensor, premise updating, future fixed points, suffering, the World OS, and the Personal OS are unified as components of a single ontological implementation architecture.

This study therefore clarifies the theoretical relationship between ontology and operating-system theory within Future Convergence Theory while extending its foundation as a theory of ontological implementation.

A Theory Correspondence

This appendix summarizes the theoretical correspondence between the present study and FCT Papers 81 and 96.

The present study does not replace either of the previous works.

Instead, it extends and integrates them by introducing the missing ontological implementation hierarchy that connects pre-judgment ontological determination with the Premise Update Operating System.

Accordingly, this appendix clarifies how the principal concepts developed in the previous papers are inherited, redefined, and integrated within the present theoretical framework.

A.1 Correspondence with FCT Paper 81

Table 1: Theoretical Correspondence with FCT Paper 81

FCT Paper 81	Present Study	Theoretical Development
Premise Updating	Updating of the Mode of Existence	The object of updating is extended from cognition to ontology.
Future Fixed Point	Convergence Structure Derived from the Mode of Existence	Reinterpreted as ontological convergence rather than a fixed objective.
Suffering	Ontological Consistency Log	Generalized from an update request log to a consistency log.
Meaning Density	Meaning Density Tensor	Redefined as an ontological implementation medium rather than a measure of semantic quantity.
World OS	Ontological Information Implementation OS	Explicitly connected to the ontological layer.
Personal OS	Mode-of-Existence Implementation OS	Reorganized as the information-processing system derived from the mode of existence.
Premise Update Operating System	Ontological Premise Update Operating System	Extended from a cognitive updating system to an ontological operating system.

A.2 Correspondence with FCT Paper 96

Table 2: Theoretical Correspondence with FCT Paper 96

FCT Paper 96	Present Study	Theoretical Development
Pre-Judgment Determination	Axiom of the Ontological Layer	The fundamental ontological condition underlying the entire theory.
Existence	Mode of Existence	Organized as existence unfolded along the temporal dimension.
Determination	Ontological Determination	The foundational structure generating the mode of existence.
World Premises	Mode of Existence	World premises are identified with the mode of existence itself.
Ontology	Ontological Layer	Defined as the foundational layer preceding system implementation.

A.3 Concepts Newly Introduced in the Present Study

The present study introduces several concepts that were not explicitly formulated in FCT Papers 81 or 96.

Table 3: Newly Introduced Concepts

Concept	Definition
Mode of Existence	The present state resulting from the temporal unfolding of ontological determination.
Ontological Updatability	The ontological property through which existence remains updateable without becoming fixed into a closed structure.
Limit of Ontological Updating	The condition in which no updateable world remains ontologically realizable within the mode of existence.
Ontological Consistency	The correspondence between the mode of existence and its informational implementation.
Ontological Consistency Log	Consistency information observed phenomenologically as suffering.
Ontological Information Medium	The Meaning Density Tensor regarded as the informational medium connecting ontology with implementation.
Ontological Premise Update Operating System	The operating system responsible for preserving Ontological Updatability.

A.4 Hierarchical Organization of the Theory

To distinguish ontology from informational implementation, the entire theoretical framework is organized into four hierarchical layers:

Ontology $\implies$ Mode of Existence $\implies$ Information Implementation $\implies$ System Implementation

Each layer performs a distinct function:

1. **Ontological Layer**

Maintains only the axiom of pre-judgment ontological determination.

2. **Mode of Existence Layer**

Represents the temporally unfolded present state of ontology.

3. **Information Implementation Layer**

Represents the mode of existence through the Meaning Density Tensor.

4. **System Implementation Layer**

Implements the World OS, Personal OS, future fixed points, cognition, judgment, and behavior.

A.5 Significance of the Theoretical Extension

FCT Paper 81 established the Premise Update Operating System.

FCT Paper 96 established the ontological principle of pre-judgment determination.

However, neither paper explicitly described the implementation hierarchy connecting ontology to operating-system implementation.

The present study introduces the following ontological implementation chain:

$$\begin{aligned} \text{Pre-Judgment Ontological Determination} &\implies \text{Mode of Existence} \\ &\implies \text{Meaning Density Tensor} \\ &\implies \text{Premise Update Operating System.} \end{aligned}$$

This implementation chain integrates ontology, informational structures, and operating-system implementation into a unified theoretical architecture.

A.6 Theoretical Position of the Present Study

The present study neither revises nor replaces FCT Papers 81 and 96.

Instead, it supplements the theoretical hierarchy that previously remained implicit, thereby enabling continuous theoretical description from ontology to operating-system implementation.

Accordingly, Future Convergence Theory may now be understood as a unified ontological implementation architecture organized according to the following hierarchy:

Ontology $\rightarrow$ Mode of Existence
 $\rightarrow$ Meaning Density Tensor
 $\rightarrow$ Premise Update Operating System
 $\rightarrow$ Future Convergence.

This correspondence clarifies the inheritance of concepts across the three studies, the redefinition of terminology, and the theoretical role of the concepts newly introduced in the present work.